



In class assignment

Python Flask – In-Class Assignment

Building Your First Flask Web Server

Objective

In this assignment, you will create a **brand-new Python Flask project** and build a basic web server with multiple pages.

You will practice:

- Creating a Python virtual environment
- Activating a virtual environment
- Installing Flask
- Creating a Flask server
- Creating multiple routes
- Serving HTML pages using `render_template()`
- Creating and submitting an HTML form
- Receiving and reading form data in Flask
- Understanding the basic flow between a browser and a Flask server

Important: This assignment will be continued and expanded during the upcoming classes. **You do NOT need to submit anything at this time.**

Part 1: Create a Brand-New Project

Create a completely new folder for this assignment.

For example:

```
flask_in_class_assignment
```

Open this folder in VS Code or your preferred code editor.

Open the terminal inside the project folder.

Part 2: Create a Virtual Environment

Create a Python virtual environment using:

```
python -m venv venv
```

This creates a virtual environment named venv. you can chose a virtual environment name of your choice ex - my_venv then the comand would be -

```
python -m venv my_venv
```

Your project should now look approximately like:

```
flask_in_class_assignment/  
└─ venv/
```

Part 3: Activate the Virtual Environment

Windows

If you are using Windows:

```
venv\Scripts\activate
```

or if your virtual env name is my_venv

```
my_venv\Scripts\activate
```

After activation, you should see something similar to:

```
(venv) or (my_venv)
```

at the beginning of your terminal prompt.

Part 4: Install Flask

With your virtual environment activated, install Flask:

```
pip install flask
```

Verify that Flask has been installed successfully:

```
pip list
```

You should see information about the installed Flask package.

Part 5: Create the Project Structure

Create the following folder and files:

```
flask_in_class_assignment/  
|  
├── venv/  
|  
├── main.py  
|  
└── templates/  
    ├── home.html  
    ├── contact.html  
    └── form.html
```

What are these files?

- `main.py` → Contains your Flask server and routes
- `templates/` → Contains the HTML pages
- `home.html` → Home page
- `contact.html` → Contact Us page

- `form.html` → Page containing the form

Flask looks for HTML templates inside a folder named:

`templates`

Part 6: Create the Flask Server

Create a file called:

`main.py`

This is where our main flask server code will be.

Part 7: Create the Home Page

Inside the `templates` folder, create:

`home.html`

Part 8: Create the Contact Us Page

Inside the `templates` folder, create:

`contact.html`

Part 9: Create the Form Page

Inside the `templates` folder, create:

`form.html`

The page should contain a form with **10 different questions**.

Part 10: Run Your Flask Server

Make sure your virtual environment is still activated.

Run:

```
python main.py
```

You should see something similar to:

```
* Running on http://127.0.0.1:5000
```

Open the URL in your browser:

```
http://127.0.0.1:5000
```

Part 11: Test All Pages

Your application should have the following routes:

Page	URL
Home	/
Contact Us	/contact
Form	/form
Form Submission	/submit-form

Test the following:

Test 1 – Home Page

Visit:

```
http://127.0.0.1:5000/
```

You should see your Home page.

Test 2 – Contact Page

Visit:

```
http://127.0.0.1:5000/contact
```

You should see your Contact Us page.

Test 3 – Form Page

Visit:

`http://127.0.0.1:5000/form`

You should see the 10-question form.

Test 4 – Submit the Form

Fill out all 10 questions and click:

Submit Form

You should see:

Form submitted successfully! Check your terminal to see the submitted data.

Look at your Flask terminal.

You should see the submitted information printed there.

Your Final Project Structure

When you are finished, your project should look like:

```
flask_in_class_assignment/  
|  
├── venv/  
|  
├── main.py  
|  
└── templates/  
    ├── home.html  
    ├── contact.html  
    └── form.html
```

Important Concepts to Understand

By the end of this exercise, you should understand the basic flow:

Browser

```
|  
| GET /
```

```
↓
Flask Server
|
| render_template()
↓
home.html
```

For the form:

```
User fills out form
|
| POST
↓
/submit-form
|
↓
Flask
|
| request.form.get()
↓
Read submitted data
|
↓
Print data in terminal
```

Pay particular attention to these Flask concepts:

```
@app.route()
render_template()
request.form.get()
```

and:

```
methods=["POST"]
```

Assignment Instructions

You are required to:

1. Start a **brand-new Flask project**.
2. Create and activate a Python virtual environment.
3. Install Flask using `pip install flask`.

4. Create a Flask server in `main.py`.
5. Create a Home page.
6. Create a Contact Us page.
7. Create a Form page.
8. Add **10 different questions** to the form.
9. Create a Flask route that receives the submitted form data in POST.
10. Read the submitted data using Flask's `request.form`.
11. Print the submitted data in the terminal.
12. Add comments to your Python and HTML code.
13. Make sure all pages and routes are working correctly.

Submission

There is no submission required at this time.

We will continue building and improving this Flask application in the upcoming class. Keep your project and make sure it is working before the next class, as we will use this same project for the next part of the assignment.

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In class assignment 4



Assignment